



National Textile University

Department of Computer Science

Subject:

Operating System

Submitted to:

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Lab no:

1

Semester:

5th

Operating Systems – COC 3071L

SE 5th A – Fall 2025

Objective

The purpose of this assignment is to:

1. Configure **Ubuntu** inside **WSL2 (Windows Subsystem for Linux v2)**.
 2. Install and configure **Git** in Ubuntu.
 3. Generate and set up **SSH keys** to connect with GitHub.
 4. Install the **C development environment** in Ubuntu.
 5. Write a **Hello World** program in `C`.
-

Part A: WSL2 & Ubuntu Setup

1. Verify WSL2 and Ubuntu installation

- Verify installation by running the following command in powershell:

```
wsl --list --verbose
```

Submit a screenshot showing Ubuntu installed and running on WSL2.

2. Update Ubuntu environment

- Run the following command in Ubuntu:

```
sudo apt update && sudo apt upgrade -y
```

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\WINDOWS\system32> wsl --list --verbose
   NAME                STATE              VERSION
*  Ubuntu-24.04         Stopped            2
PS C:\WINDOWS\system32>
```

Part B: Git & GitHub SSH Setup

1. Configure Git

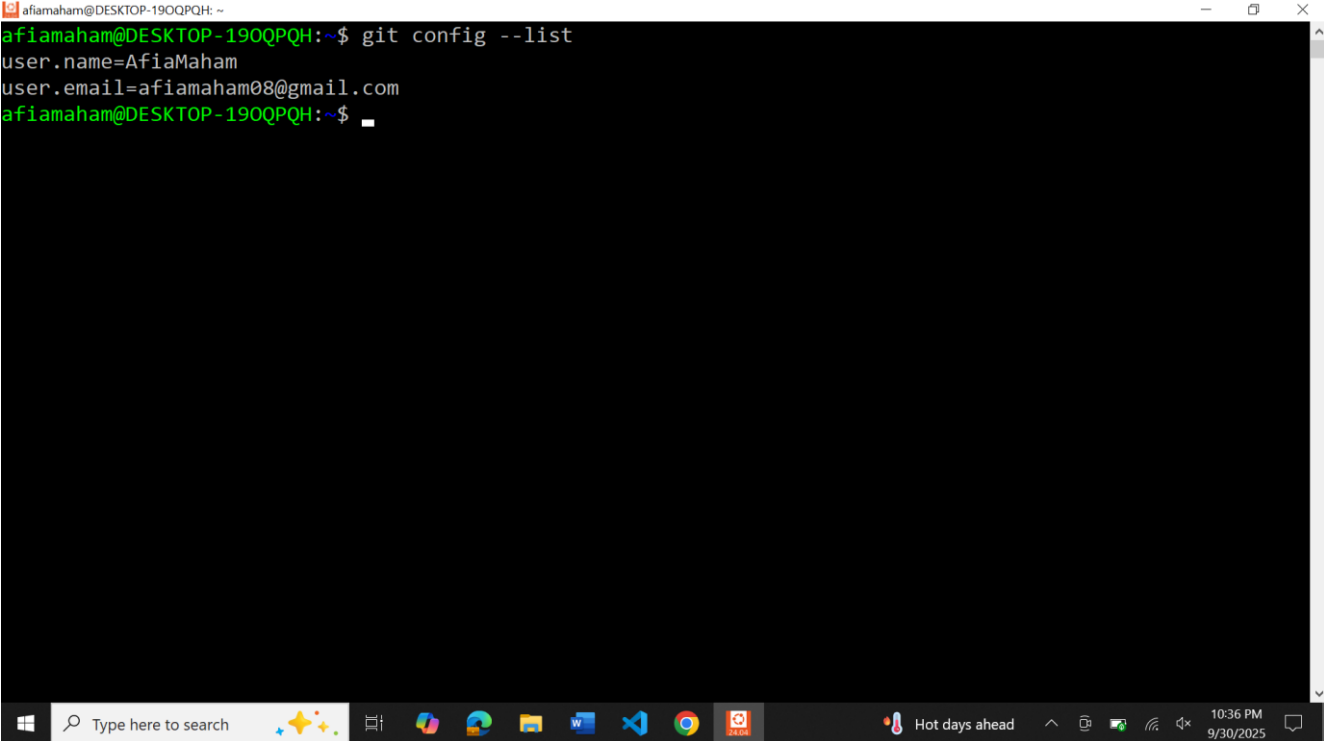
- Set your name and email:

```
git config --global user.name "Your Name"
git config --global user.email "your@email.com"
```

- Show your config:

```
git config --list
```

Submit a screenshot.

A screenshot of a Windows terminal window. The title bar shows 'afiamaham@DESKTOP-190QPQH: ~'. The terminal content shows the command 'git config --list' being executed, with the output: 'user.name=AfiaMaham' and 'user.email=afiamaham08@gmail.com'. The prompt 'afiamaham@DESKTOP-190QPQH:~\$' is visible at the bottom. The Windows taskbar is at the bottom, showing the search bar, task view, and various application icons. The system tray on the right shows 'Hot days ahead' and the date/time '10:36 PM 9/30/2025'.

2. Generate SSH Keys

- Run:

```
ssh-keygen -t ed25519
```

- Copy the public key:

```
cat ~/.ssh/id_ed25519.pub
```

- Add this key to your GitHub account under **Settings** → **SSH and GPG keys**.

3. Test Connection

```
ssh -T git@github.com
```

- Submit a screenshot showing successful authentication.

```
afiamaham@DESKTOP-190QPQH: ~  
afiamaham@DESKTOP-190QPQH:~$ ssh -T git@github.com  
Hi AfiaMaham! You've successfully authenticated, but GitHub does not provide shell access.  
afiamaham@DESKTOP-190QPQH:~$
```

Part C: C Programming Environment & Practice

Step 1: Install Build Tools

Before writing C programs, install the **build-essential** package which contains `gcc` , `g++` , and other tools required for compiling.

Run:

```
sudo apt install build-essential
```

Verify installation by checking the version of `gcc` :

```
gcc --version
```

- Submit a screenshot of successful installation and version output.

```
afiamaham@DESKTOP-190QPQH: ~  
afiamaham@DESKTOP-190QPQH:~$ gcc --version  
gcc (Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0  
Copyright (C) 2023 Free Software Foundation, Inc.  
This is free software; see the source for copying conditions. There is NO  
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  
afiamaham@DESKTOP-190QPQH:~$
```

Step 2: How to run a C Program

1. First write a C program in a file with `.c` extension.
2. Compile the file using `gcc filename.c -o filename.out`
3. Execute it using `./filename.out`

Breakdown

- ♦ `gcc`
 - ♦ This is the GNU Compiler Collection command.

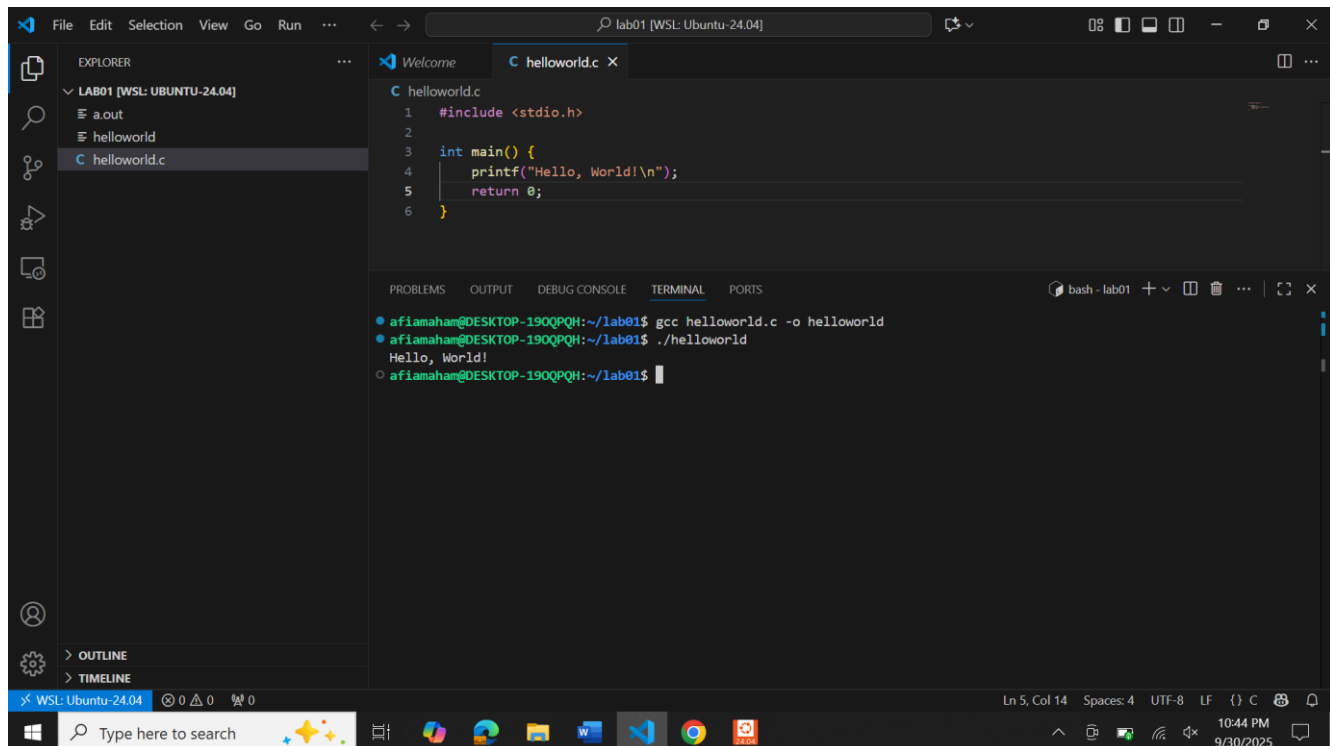
- ♦ It compiles C (and other languages like C++) programs into machine code that can be executed by the computer.
 - ♦ `filename.c`
 - ♦ This is the source code file you wrote in C.
 - ♦ Example: `hello.c` contains your C program.
 - ♦ `-o filename.out`
 - ♦ The option `-o` means “output.”
 - ♦ By default, `gcc` creates an executable file named `a.out` if you don't specify anything.
 - ♦ With `-o`, you can choose the name of the output executable.
 - ♦ In this case, the compiled file will be named `filename.out`.
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Step 3: Write a C Program

Write a simple C program of your choice. It can be a **Hello World** program or any other.

Submission: For the program, submit:

- ♦ The C source code (`.c` file).
- ♦ Screenshot of execution (`./program`)



The screenshot shows the Visual Studio Code interface with a C program named `helloworld.c` open. The code is as follows:

```
1 #include <stdio.h>
2
3 int main() {
4     printf("Hello, World!\n");
5     return 0;
6 }
```

The terminal window at the bottom shows the execution of the program:

```
afiamahan@DESKTOP-190QPQH:~/lab01$ gcc helloworld.c -o helloworld
afiamahan@DESKTOP-190QPQH:~/lab01$ ./helloworld
Hello, World!
afiamahan@DESKTOP-190QPQH:~/lab01$
```

The Explorer panel on the left shows the file structure with `helloworld.c` selected. The status bar at the bottom indicates the file is at line 5, column 14, with 4 spaces, in UTF-8 encoding, and the system clock shows 10:44 PM on 9/30/2025.

Deadline

- ♦ Submit before **12:00 AM, 25 September, 2025.**